

8 July 2020

① $\exists$ lin. homog. d.e. with c.c. of order 7 that has as sol. the fct:

a) $t^2 \cos 2t, (t \sin 2t), e^{7t}$

b) $t^2 \cos 2t, t \sin 7t, e^t$

c) $t^2 \cos 2t$

$t^2 \cos 2t$ sol. $\Leftrightarrow \pm 2i$ root of multiplicity 3 (triple root)

$t \sin 2t$ sol. $\Leftrightarrow \pm 2i$ root of multiplicity 2 (double root)

e^{7t} sol. $\Leftrightarrow 7$ root of multiplicity 1 (simple root)

e^t sol. $\Leftrightarrow 1$ " " " " —

a) char. equation has at least degree $3 \times 2 + 1 = 7$

$\Rightarrow$ yes $\exists$.

b) at least degree: $3 \times 2 + 1 + 2 \times 2 + 1 = 11 > 7 \Rightarrow$ no, $\nexists$.

c) at least degree: $3 \times 2 = 6 < 7 \Rightarrow$ yes, $\exists$.

② a) $x = A_0 \cos(3t - \phi_0)$, $A_0 \geq 0$, $\phi_0 \in [0, 2\pi] \rightarrow$ gen. sol. of d.e. $x'' + 9x = 0$?

CHAR. EQ. METHOD:

$$x'' + 9x = 0$$

$$x^2 + 9 = 0 \Rightarrow x = \pm 3i \rightarrow \cos 3t, \sin 3t.$$

$$x = c_1 \cos 3t + c_2 \sin 3t; c_1, c_2 \in \mathbb{R}.$$

$$x = A_0 \cos(3t - \phi_0) = A_0 (\cos 3t \cdot \cos \phi_0 + \sin 3t \cdot \sin \phi_0) =$$

$$= (A_0 \cos \phi_0) \cos 3t + (A_0 \sin \phi_0) \sin 3t = c_1 \cos 3t + c_2 \sin 3t.$$

$$(\forall) (c_1, c_2) \in \mathbb{R}^2 \setminus \{0, 0\}, \exists! (A_0, \phi_0) \in (0, \infty) \times [0, 2\pi] \text{ s.t. } \begin{cases} c_1 = A_0 \cos \phi_0 \\ c_2 = A_0 \sin \phi_0 \end{cases}$$

polar coordinates of (c_1, c_2) .

for $(c_1, c_2) = (0, 0) \Rightarrow A_0 = 0. \Rightarrow$ yes, gen. sol.

b) $A_1 \in \mathbb{R}$. Particular sol. of $x'' + 9x = A_1 \cos 3t$, form $x_p = a \cdot t \cdot \sin 3t$.

$$x'_p = a \cdot \sin 3t + 3a \cdot t \cdot \cos 3t$$

$$x''_p = 3a \cdot \cos 3t + 3a \cdot \cos 3t - 9at \cdot \sin 3t$$

Replace: $6a \cdot \cos 3t - 9at \cdot \sin 3t + 9(a \cdot t \cdot \sin 3t) = \overset{A_1}{6a} \cdot \cos 3t, (\forall) t \in \mathbb{R}.$

$$\Rightarrow 6a \cdot \cos 3t = A_1 \cdot \cos 3t$$

$$\Rightarrow 6a = A_1 \Rightarrow a = \frac{A_1}{6}$$

$$x_p = \frac{A_1}{6} \cdot t \cdot \sin 3t$$

c) IVP:
$$\begin{cases} x'' + 9x = A_1 \cos 3t \\ x(0) = 0 \\ x'(0) = 0 \end{cases} \quad \text{sol. ?}$$

Gen. sol. from b) $\Rightarrow \theta(t) = \frac{A_1}{6} t \cdot \sin 3t$

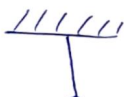
$\theta(0) = 0 \quad \checkmark$

$\theta'(0) = \dots = 0 \quad \checkmark$

This IVP has a unique sol. by the Existence and Uniqueness Theorem.
(second order lin. d.e. w/ 2 im. cond.)

$\Rightarrow$ unique sol: $\theta(t) = \frac{A_1}{6} t \cdot \sin 3t$

Motion of a simple pendulum when $\theta(t)$ = measure in radians of $\nearrow$ rod, vert.

$\theta(0) = 0$
 $\theta'(0) = 0$ $\} \Rightarrow$ initial position: 

? $\theta(t) = 0 \Leftrightarrow t \cdot \sin 3t = 0$

$\Rightarrow \sin 3t = 0$

$\Rightarrow 3t = k\pi \Rightarrow t = \frac{k\pi}{3}$

$\Rightarrow \theta\left(\frac{k\pi}{3}\right) = 0, (\forall) k \in \mathbb{Z}$

θ changes sign around $\frac{k\pi}{3}$

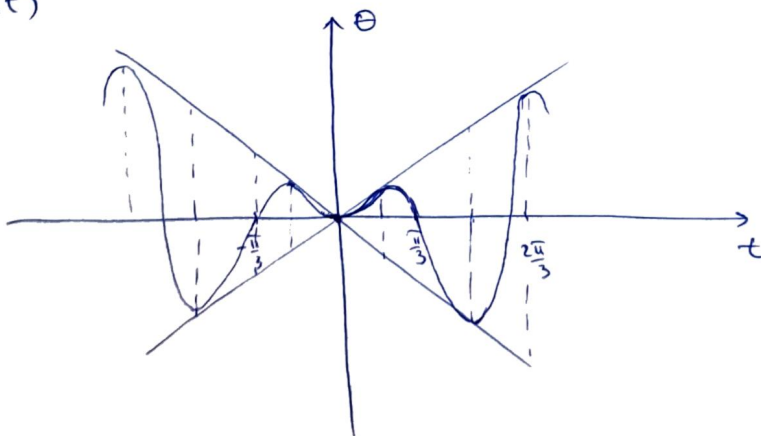
$\Rightarrow$ pendulum oscillates around $\theta^* = 0$.

$\lim_{t \rightarrow \infty} |\theta(t)| = \lim_{t \rightarrow \infty} \left| \frac{A_1}{6} t \cdot \sin 3t \right| = \infty \Rightarrow$ amplitude is increasing.

$\Rightarrow$ not possible $\Rightarrow$ RESONANCE.

$\theta(-t) = \theta(t), (\forall) t \geq 0$.

Graph: $A_1 = 6$, fixed.
of $\theta(t)$



) $A_2 \in \mathbb{R}$; part. sol. of $x'' + 9x = A_2$

$$x_p = \frac{A_2}{9}$$

$$9 \cdot \frac{A_2}{9} = A_2 \checkmark$$

e) part. sol. of $x'' + 9x = \underline{A_1 \cdot \cos 3t} + \underline{A_2}$.

$$x_{p1} = \frac{A_1}{6} t \sin 3t \quad \text{sol. of } x'' + 9x = \underline{A_1 \cdot \cos 3t}$$

$$x_{p2} = \frac{A_2}{9} \quad \text{sol. of } x'' + 9x = \underline{A_2}$$

SUPERPOSITION PRINCIPLE:

$$x_p = \frac{A_2}{9} + \frac{A_1}{6} t \sin 3t.$$

③a) Sol. of IVP. $y' = y$, $y(0) = 1$

$$y(x) = e^x$$

b) Euler's numerical formula with $h = 0.01$ to approx. sol. of this IVP in $[0, 1]$.

$$\boxed{x_k = k \cdot h} = k \cdot 0.01, \quad k = \overline{0, 100}$$

$$\boxed{y_{k+1} = y_k + h \cdot y_k} = (1+h)y_k = 1.01 \cdot y_k, \quad k = \overline{0, 100}.$$

$$\begin{aligned} x_0 &= 0 \\ y_0 &= 1 \end{aligned}$$

c) find a rational approx. of e (Euler's ct.)

$$x_{100} = 100 \cdot 0.01 = 100 \cdot \frac{1}{100} = 1$$

$$f(1) = e^1 = e = e^{x_{100}}$$

? $\Rightarrow \underline{y_{100}}$ = the approx. given by b) for e .

$$\checkmark y_{k+1} = 1.01 \cdot y_k, \quad \underline{y_0 = 1}.$$

$$\Rightarrow y_k = (1.01)^k, \quad \forall k \geq 0.$$

$$\Rightarrow y_{100} = \left(\frac{101}{100}\right)^{100} \rightarrow \text{rational nr.}$$

$$y_0 = 1$$

$$y_1 = 1.01 \cdot y_0 = 1.01 \cdot 1$$

$$y_2 = 1.01 \cdot y_1 = 1.01 \cdot 1.01 \cdot 1 = (1.01)^2$$

$\vdots$

$$y_k = (1.01)^k$$

$\frac{1}{2}$
 $c = 0.57 \rightarrow$ Euler-Mascheroni
 $e = 2.71 \rightarrow$ Euler

4. a) planar system:

$$\begin{cases} \dot{x} = -y\sqrt{3} + x(9-x^2-3y^2) \\ \dot{y} = \frac{x}{\sqrt{3}} + y(9-x^2-3y^2) \end{cases}$$

a) type? stability of eq. (0,0) using lin. method.

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x,y) = \begin{pmatrix} -y\sqrt{3} + 9x - x^3 - 3xy^2 \\ \frac{x}{\sqrt{3}} + 9y - yx^2 - 3y^3 \end{pmatrix}$
 $f \Rightarrow$ polynomial function.
 $\Rightarrow f \in C^1(\mathbb{R})$.

$$Jf(x,y) = \begin{pmatrix} 9-3y^2-3x^2 & -\sqrt{3}-6xy \\ \frac{1}{\sqrt{3}}-2xy & 9-x^2-9y^2 \end{pmatrix}$$

$$Jf(0,0) = \begin{pmatrix} 9 & -\sqrt{3} \\ \frac{1}{\sqrt{3}} & 9 \end{pmatrix}$$

$$\begin{vmatrix} 9-\lambda & -\sqrt{3} \\ \frac{1}{\sqrt{3}} & 9-\lambda \end{vmatrix} = 0 \Rightarrow (9-\lambda)^2 + 1 = 0 \Rightarrow (9-\lambda)^2 = -1 \Rightarrow \\ \Rightarrow (9-\lambda) = \pm i \Rightarrow \lambda_{1,2} = 9 \pm i$$

$\text{Re}(\lambda_1) = \text{Re}(\lambda_2) = 9 \neq 0 \Rightarrow$ eq. (0,0) is HYPERBOLIC
 $\Rightarrow$ LINEARIZATION METHOD WORKS.

$$\begin{aligned} \dot{x} &= Jf(x^*)(x-x^*) = \begin{pmatrix} 9 & -\sqrt{3} \\ \frac{1}{\sqrt{3}} & 9 \end{pmatrix} (x-x^*) = \begin{pmatrix} \dots \\ \dots \end{pmatrix} \begin{pmatrix} x-0 \\ y-0 \end{pmatrix} = \\ &= \begin{pmatrix} 9x - \sqrt{3}y \\ \frac{1}{\sqrt{3}}x + 9y \end{pmatrix} \Rightarrow \begin{cases} \dot{x} = 9x - \sqrt{3}y \\ \dot{y} = \frac{x}{\sqrt{3}} + 9y \end{cases} \end{aligned}$$

$\lambda_{1,2} = 9 \pm i \in \mathbb{C}$, $\text{Re}(\lambda_1) = \text{Re}(\lambda_2) = 9 \neq 0 \Rightarrow$ FOCUS.

$> 0 \Rightarrow$ global repeller.
 b) $p(t, 3, 0) = (3 \cos t, \sqrt{3} \sin t)$, $(\forall) t \in \mathbb{R} \rightarrow$ orbit? slope?

$p(t, 3, 0) =$ sol. of the sys. that also satisfies $x(0)=3$, $y(0)=0$.

Let $x(t) = 3 \cos t$, $y(t) = \sqrt{3} \sin t$

$$x(0) = 3 \quad \checkmark$$

$$y(0) = 0 \quad \checkmark$$

$$= -3 \sin t + 27 \cos t - 27 \cos^3 t - 27 \sin^3 t \\ = -3 \sin t + 27 \cos t - 27$$

$$x'(t) = -3 \sin t = -\sqrt{3} \cdot \sqrt{3} \sin t + 3 \cos t (9 - 9 \cos^2 t - 3 \cdot 3 \sin^2 t)$$

$$y'(t) = \sqrt{3} \cos t = \frac{3 \cos t}{\sqrt{3}} + \sqrt{3} \sin t (9 - 9 \cos^2 t - 3 \cdot 3 \sin^2 t), \quad t \in \mathbb{R}$$

$\Rightarrow \dots$

$\Rightarrow$ valid $\Rightarrow p(t, 3, 0) = (3 \cos t, \sqrt{3} \sin t)$

orbit: $\Gamma = \left\{ \underbrace{(3\cos t)}_x, \underbrace{(\sqrt{3}\sin t)}_y : t \in \mathbb{R} \right\}$

$$3\cos t = x \Rightarrow \cos t = \frac{x}{3}$$

$$\sqrt{3}\sin t = y \Rightarrow \sin t = \frac{y}{\sqrt{3}}$$

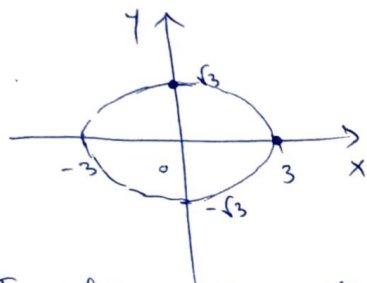
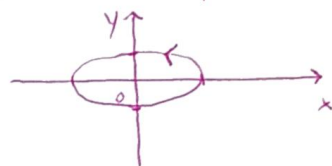
$$\Rightarrow \left(\frac{x}{3}\right)^2 + \left(\frac{y}{\sqrt{3}}\right)^2 = 1 \Rightarrow \text{ellipse.}$$

$$\gamma_{(3,0)}^y = \{p(t, 3, 0) : t \in \mathbb{R}\} = \{(3\cos t, \sqrt{3}\sin t) : t \in \mathbb{R}\}$$

↳ ORBIT: the parametric curve of parametric eq

$$\begin{cases} x = 3\cos t \\ y = \sqrt{3}\sin t \end{cases} t \in \mathbb{R}$$

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{\sqrt{3}}\right)^2 = 1 \text{ (ellipse)}$$



$$t=0, x(0)=3, y(0)=0$$

$$t=\frac{\pi}{2}, x\left(\frac{\pi}{2}\right)=0, y\left(\frac{\pi}{2}\right)=\sqrt{3}$$

c) Transform given system to coord. $(r, \varphi) \in [0, \infty) \times [0, 2\pi)$ related to the cartesian coord. (x, y) by $\frac{x}{\sqrt{3}} = r \cos \varphi$

$$y = r \sin \varphi$$

$$\left(\frac{x}{\sqrt{3}}\right)^2 + y^2 = r^2, \quad \tan \varphi = \frac{y}{\frac{x}{\sqrt{3}}} = \frac{\sqrt{3}y}{x}$$

$$\Rightarrow \begin{cases} x^2 + 3y^2 = 3r^2 \rightarrow ' \Rightarrow 2 \cdot x \cdot \dot{x} + 3 \cdot 2 \cdot y \cdot \dot{y} = 3 \cdot 2 \cdot r \cdot \dot{r} \quad | :2 \\ \tan \varphi = \sqrt{3} \cdot \frac{y}{x} \quad | ' \Rightarrow x \dot{x} + 3y \dot{y} = 3r \dot{r} \end{cases}$$

$$\Rightarrow (1 + \tan^2 \varphi) \dot{\varphi} = \sqrt{3} \cdot \frac{\dot{y} \cdot x - y \cdot \dot{x}}{x^2}$$

$$\frac{\dot{\varphi}}{\cos^2 \varphi} = \sqrt{3} \cdot \frac{\dot{y}x - y\dot{x}}{x^2}$$

$$\begin{cases} 3r\dot{r} = x(\dots) + 3y(\dots) \\ \frac{\dot{\varphi}}{\cos^2 \varphi} = \sqrt{3} \cdot \frac{x(\dots) - y(\dots)}{x^2} = \dots \end{cases} \quad (\Rightarrow)$$

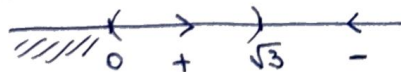
$$\begin{cases} 3r\dot{r} = 3r^2 \cdot (3 - 3r^2) \\ \frac{\dot{\varphi}}{\cos^2 \varphi} = \frac{3r^2}{3r^2 \cdot \cos^2 \varphi} \end{cases} \quad (c)$$

$$(c) \begin{cases} \dot{r} = r(3 - 3r^2) \leftarrow \text{to } r \\ \dot{\varphi} = 1 \end{cases}$$

d) Phase portrait? $r \in [0, \infty), \varphi \in [0, 2\pi)$.

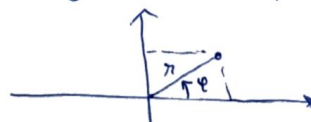
UNCOUPLED SYSTEM in coord. (r, φ) : $\begin{cases} \dot{r} = r(3 - 3r^2) = r(3 - r\sqrt{3})(3 + r\sqrt{3}) \\ \dot{\varphi} = 1 > 0 \Rightarrow \text{trigonometric sense} \end{cases}$

r :



$(r, \varphi) \rightarrow$ like polar, but a bit deformed
 r = dist. to origin
 φ = angle of (x, y)

geometric interpretation of polar coord.



After motivation, we could also

equil. point $r^* = \frac{1}{\sqrt{3}} \Rightarrow$ orbit Γ d b)

$r \in (0, \frac{1}{\sqrt{3}}) \Rightarrow$ point inside Γ

$r > \frac{1}{\sqrt{3}} \Rightarrow$ point outside Γ .

Reading the phase portrait of $\dot{r} = r(9-3r^2) \Rightarrow$ an orbit that starts inside Γ stays inside Γ for any time, departing from origin, and approaching Γ as $t \rightarrow \infty$; an orbit that starts outside Γ stays outside Γ for any time, approaching Γ as $t \rightarrow \infty$.

